Reyshwanth Ganeshan

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EDUCATION

TEXAS A&M UNIVERSITY 2023-2027
Ph.D. Mechanical Engineering GPA: 3.5*/4.0
Advisor: Dr.Sivakumar Rathinam

TEXAS A&M UNIVERSITY 2021-2023
M.S. (Thesis) Mechanical Engineering
GPA: 4.0/4.0

IIT MADRAS 2017-2021
B.Tech. Mechanical Engineering

SKILLS

PROGRAMMING
C++, Python, ROS/ROS2, OpenCV, Julia, Docker, PostgreSQL, QGIS, CPLEX, Gurobi, Git

ML & DATA
Pytorch, Tensorflow, CUDA, SQL, Stable-Baselines3, Gymnasium

SIMULATION
Gazebo, Unity, IsaacSim, MATLAB

SELECTED COURSES

Convex Optimization, **Deep Reinforcement Learning**, **Decision Making for Autonomous Systems**, **Machine Learning**, Analysis of Algorithms, Intelligent Transportation Systems, Design of Non Linear Control Systems, Control Systems Design, Statistics in Research

PUBLICATIONS

- **R. Ganeshan**, et al., (2026). Bi-Objective Dynamic Zone Navigation*AIAA SCITECH 2026 Forum*
- N. Kumar, **R. Ganeshan**, et al., (2026). Coordination and Planning Algorithms for a UAS Fleet in GPS-Denied Environments *AIAA SCITECH 2026*
- **R. Ganeshan**, et al., (2025). Multi-Domain UGV-UAV Team Path Planning. *AIAA SCITECH 2025 Forum*
- **R. Ganeshan**, S. Darbha. (2024). A Diagnostic Machine Learning Algorithm for Air Brakes in Commercial Vehicles. *IFAC-PapersOnLine*

ACHIEVEMENTS

- Led an **autonomous car competition** team of 35 to consecutive **2nd Place** finishes Autodrive Challenge II(2024,2025)
- Co-led an **FSAE** team of 45 and declared **National Champions** amongst 60 teams at Formula Bharat 2020
- Emerged in the top **0.16%** of **IIT-JEE Advanced 2017** Exam amongst **1.2 million candidates**

PROFESSIONAL EXPERIENCE

TEAM CAPTAIN Aug 2023 - Present
GM SAE Autodrive Challenge II
- Deployed **PointPillar** based Multi Class (Deer, Barrell and Barricade) Detection with fast inference times of **80ms**, fusing it with Yolov8 detection from camera
- Integrated multi-modal **perception stack** with **LiDAR, Radar** and camera for object detection, tracking, and adverse weather handling (fog, glare)
- Developed and deployed **L4 autonomous vehicle** with integrated **IMU, GNSS, wheel encoder,Cameras and LiDAR** for robust **pose estimation**
- Implemented **LiDAR SLAM** algorithms and **probabilistic filtering** for **GPS-denied localization** and real-time mapping in urban environments
- Built **trajectory optimization** algorithms using constraint modified **MPC controllers** for safe obstacle avoidance and autonomous navigation

GRADUATE RESEARCH ASSISTANT Aug 2023 - Present
Autonomous Systems Lab
- Integrated Applied Research in **robotics, AI and multi-agent navigation**
- Developed **multi-agent** navigation algorithms for **multi-objective** applications using heterogenous agents, with **real-world demos** using **UAV-UGV** teams
- Compared use of DeepRL as well as conventional methods (A*, Dijkstra etc.) to solve multi-agent routing in GPS denied environments, with outdoor demos

VEHICLE SOFTWARE INTERN Aug 2022 - Dec 2022
Tesla Inc.
- Built **HIL Firmware Regression Testers** for all Tesla Vehicles(S,3,X,Y and more)
- Set up firmware testing on Python over **CAN** while performing safety critical maneuvers such as ABS braking improving testing speed by over **60%**

SELECTED PROJECTS

GPS DENIED MULTI AGENT COVERAGE Sept 2025- Current
Texas A&M University
- Set up and demonstrated **outdoor GPS-denied** landmark-based coordination algorithms for a heterogenous team of UAVs and UGVs
- Used **MAPPO** with attention augmented **Actor-Critic** on custom GPS-Denied coverage environments for **Persistent Target Coverage** with **Fixed Wing Agents**
- Implemented a novel, robust and fast **convex optimization based** state estimation using landmark range information only - replacing traditional EKF methods

BI-OBJECTIVE DYNAMIC ZONE AVOIDANCE Jan 2025 - June 2025
Texas A&M University
- Implemented and improved **graph/search-based motion planning** using **BOA* (Bi-Objective A*)** for dynamic zone navigation using **fixed wing aircrafts**
- Formulated **cost functions and optimization frameworks** balancing **risk vs traversal time** in trajectory selection
- Developed **geometry-based path reasoning** with **grid-based planners** handling **mobile, dynamic, shape-changing zones** and **time-varying risk fields**
- Designed novel **cost functions** using **proximity-based risk scoring mechanisms** for safe trajectory planning
- Improved algorithm **convergence speed** by 72% through enhanced **exploration strategies** in complex environments

MULTI-DOMAIN UAV-UGV TEAM Aug 2024 - Dec 2024
Texas A&M University
- Developed and deployed a UAV-UGV team using **GPLA*** (Generalized Permanaent Labelling algorithm) for Search and Rescue Applications
- Implemented costmap based **dynamic obstacle avoidance** and re-planning on Clearpath Huskys using LiDAR and nav2 software stacks
- Validated the autonomous stack in **Gazebo**, demonstrating multi-agent coordination and obstacle clearing using **UR5 Arm** attached to the UGV
- Extended the scope of the algorithm by adding **online replanning** with dynamic graph updates, enabling real world deployment